NAVIGATION

Finding your way with a paper map and a compass when phones, GPS or road signs can't be counted on.

DOCUMENT TR-04
SERIES TRAINING
REVISION B // NOV 2026
STATUS PUBLIC ISSUE

GUIDE / VERIFY: General preparedness information, not a substitute for professional training, local rules, manufacturer instructions or medical care. **Practice skills in safe conditions before you need them, and verify important information with current official sources.**

Phones and car GPS work well until the battery dies, the cell network is down, or the road on the screen is closed. A paper map and a basic compass never need charging. Keep a road map of your area in each car and go-bag, with your evacuation routes marked.

1. READING A MAP

MAP PART	WHAT IT TELLS YOU
Legend	What each symbol and color means: road types, rivers, buildings, trails, boundaries.
Scale	How map distance matches real distance. On a 1:24,000 map (a standard USGS topographic map), 1 inch = 2,000 feet. Use the scale bar to measure.
North arrow	Which way is north on the map. Most maps put north at the top. Many show the **declination** too (see section 3).
Contour lines	On topographic maps, each line joins points at the same height. Lines close together = steep; far apart = flat. A V pointing uphill = a stream valley.
Grid	Lines that let you describe a location precisely (latitude/longitude, or UTM grid on topographic maps).

2. THE COMPASS

PARTS OF A BASEPLATE COMPASS

- **Needle:** the red end points to magnetic north.
- **Housing (bezel):** the dial marked 0 to 360 degrees, with an orienting arrow and lines inside.
- **Direction-of-travel arrow:** on the baseplate, points where you'll walk.
- **Index line:** where you read the bearing off the dial.

KEEP IT ACCURATE

- Hold it flat and level, at waist height.
- Keep it away from steel and magnets: car bodies, belt buckles, phones, power lines.
- Directions: 0/360 = north, 90 = east, 180 = south, 270 = west.

3. DECLINATION: MAP NORTH VS. COMPASS NORTH

A compass points to **magnetic** north; maps use **true** north. The difference, called declination, depends on where you are and changes slowly over the years. In parts of the U.S. it is more than 10 degrees: over a mile, a 10-degree error puts you nearly 1,000 feet off.

- Look up your current declination with NOAA's free online calculator, and write it below.
- Many compasses let you set declination once, so you don't have to calculate it.
- If not: with **east** declination, subtract it from map bearings to get compass bearings. With **west** declination, add it.
- Memory aid: "East is least, west is best" (for map-to-compass).

4. THREE THINGS TO PRACTICE

SKILL	HOW
Orient the map	Lay the map flat. Put the compass on it with the dial set to 0 (north) and the baseplate edge along a north-south line. Turn the map and compass together until the red needle sits inside the orienting arrow ("red in the shed"). The map now matches the land around you.
Follow a bearing	Turn the dial until the bearing you want sits at the index line. Hold the compass flat and turn your whole body until the red needle is in the shed. Walk where the travel arrow points. Pick a landmark on that line, walk to it, repeat.
Find where you are	Take bearings to two or three landmarks you can see and find on the map. Draw each bearing line back from its landmark on the map. You are where the lines cross.

5. WITHOUT A COMPASS, AND STAYING FOUND

- The sun rises roughly in the east and sets roughly in the west. At midday in the U.S. it is to the south.
- Use **handrails**: roads, rivers, power lines and ridges you can follow.
- Know your **backstop**: a road or river that tells you you've gone too far.
- **Pace count:** count every time your left foot lands. Measure how many it takes you to walk 100 yards on flat ground. Write it below.
- Tell someone where you're going and when you'll be back.
- Look back often: the way home looks different from the other direction.
- If you're lost: **stop**. Stay calm, think, look at the map, and plan before moving. If you can't work it out, stay put, stay warm and signal (three whistle blasts).
- Download offline maps on your phone too, and carry a power bank. Use both.

6. MY NAVIGATION CARD

Declination here: _____ **Evacuation route 1:** _____

Checked on (date): _____ **Evacuation route 2:** _____

My paces per 100 yards: _____ **Meeting place:** _____

DATE	PLACE	SKILL PRACTICED	NOTES

SOURCES / FURTHER REFERENCE

1. U.S. Geological Survey: Topographic Map Symbols; The National Map. usgs.gov
2. NOAA National Centers for Environmental Information: Magnetic Field Calculators (declination). ngdc.noaa.gov/geomag/calculators
3. National Park Service: Hiking Safety; The Ten Essentials. nps.gov
4. FEMA: Ready.gov: Evacuation. ready.gov/evacuation

IMPORTANT: Declination changes over time; check yours every few years. Practice in a familiar place before relying on these skills.